Candidate No : _______________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Computer Applications (BCA)

Semester-II University Examination May-2018

CA118 Advanced Programming

Date: 11/05/2018, Friday Time: 10.00 am to 1.00 pm Maximum Marks : 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **Figure to right indicates marks.**
3. **Make suitable assumption when necessary.**

SECTION – 1

Q.1 Do as directed **[11]**

1. A data structure that can store related information of different data types together is
   (a) Array (b) String (c) Structure (d) All of these
2. A structure can be placed inside another structure and is known as____
3. Size of a union is determined by size of the________
   (a) First member in the union (b) Last member in the union
   (c) Biggest member in the union (d) Sum of the sizes of all members
4. A structure member variable is generally accessed using the _____ operator.
5. Which of the following is not a file opening mode?
   (a) a (b) ab (c) ab+ (d) abc
6. When fopen() is not able to open a file, it returns______.
7. ________file opening mode keeps old contents and allows write into the file.
8. What is the output of following code segment:
   int no=1001; int *ptr=&no;
   printf("%d",*ptr);
9. What is the output of following code segment:
   int array[10]={20,30,40,50};
   printf("%d", *arr+2);
10. What is void pointer?
11. If unsigned integer occupies 2 bytes then what is the output of the following code segment:
    float no=30; float * fptr=&no;
    printf("%d",sizeof(fptr));

Q.2

(A) What is pointer? Explain how to declare & use a pointer. What is NULL pointer? **[06]**

(B) What is a structure? What is structure tag? Explain how a structure variable can be declared and how can you access the members of a structure by using example. **[06]**

OR

Q.2

(A) Explain about union with suitable example. Differentiate Structures and Unions. **[06]**

(B) Explain getc() and putc(), fgets() and fputs(), getw() and putw() with reference to file handling. **[06]**

Q.3

(A) Explain malloc, calloc and realloc functions with example. **[06]**

(B) Explain any six file opening modes with all details. **[06]**

OR

Q.3

(A) How can you use structures inside a function? Explain with example. **[06]**

(B) Explain fopen(), fseek() and fwrite() functions with example **[06]**

SECTION – 2

Q.4 Answer the questions **[11]**

1. A linear collection of data elements where the linear node is given by means of pointer is called_____
 (a) Linked list (b) Node list (c) Primitive list (d) None
2. A linked list node contains at least _________ fields.
3. The operation of processing each element in the list is known as
 (a) Sorting (b) Merging (c) Inserting (d) Traversal
4. A variant of linked list in which last node of the list points to the first node of the list is___
 (a)Singly linked list (b) Doubly linked list (c) Circular linked list (d) Multiply linked list
5. In stack if TOP=_____________ then it is underflow condition.
6. In queue, insertion is done at ___________end.
 (a) Front (b) Rear (c) First (d) None of these
7. If the capacity of stack is given as N. When TOP = ________then it is overflow condition in PUSH operation of stack.
8. When F =________then it is underflow in queue delete operation.
9. When Top=_______then there is one element in the stack.
10. What is the output of following code segment:

```
 #define f(x) x*x
 #include<stdio.h>
void main() {
          printf("%d",f(3-2)); }
```

11. In queue when F=R and both are >=0. How many elements are there in a queue?

Q.5

(A) What is a linked list? Differentiate between arrays and linked lists. **[06]**

(B) What is preprocessor? Explain about following preprocessor directives with example. **[06]**
(1) #define (2)#include

OR

Q.5

(A) Write an algorithm to create a singly linked list. **[06]**

(B) Convert following expressions from infix to postfix: **[06]**
(a) m +n *p / q (b) w + (x – y) * z

Q.6

(A) Write algorithm for following operations on stack: **[06]**
(a) PUSH (b) POP

(B) Write algorithm for delete operation in Queue. **[06]**

OR

Q.6

(A) Write an algorithm to insert an element in the list at the given position. **[06]**

(B) A queue Q is given with F=0, R=1. Capacity (N) of queue 4. The elements of queue are given as 25, 34. Here, front (F) is pointing towards 25 and rear (R) is pointing towards 34. Show the contents of queue when following operations take place: **[06]**

(1) INSERT (Q,100) (2) INSERT(Q,200) (3) DELETE(Q) (4) INSERT(Q,500) (5) DELETE(Q) (6) DELETE(Q)